

Research Paper

Optimizing biofuel production from algae using four-element framework: Insights for maximum economic returns

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ABSTRACT

This paper introduces a Four Element Strategy (FES) focused on concentration of CO₂ gas (C_{c,g}), energy input to the process, concentration of nutrients, and the type of algal strain to maximize biofuel production (BY_p) from algae strains and evaluate associated economic benefits. Two algal strains encompassing slain algae *Nannochloropsis* sp. (Ns), and freshwater strain *Chlorella vulgaris* (Cv) were used in the current study. Response Surface Methodology (RSM) and Box Behnkin Design (BBD) are integrated with a biokinetic model to explore the relationships among growth parameters in the ranges; C_{c,g} ~ 0.03–20 %, Nitrogen (N) to phosphorus (P) ratio (R_{N/P}) ~ 1–11, and light intensity (L_i) ~ 100–400 μE m⁻² s⁻¹ with corresponding CO₂ bio-fixation rate (R_{CO₂}, g L⁻¹ d⁻¹), biomass accumulation (X, g L⁻¹), and algal growth rate (μ, d⁻¹). Optimization results indicate that the most productive conditions involve L_i in the range of 100–270 μE m⁻² s⁻¹, C_{c,g} between 0.03 % and 15 %, and R_{N/P} of 3.5–11, achieving maximum R_{CO₂}, X, and μ at 1 g L⁻¹ d⁻¹, 6 g L⁻¹, and 1.6 d⁻¹, respectively. Further optimization using the desirability function supports achieving R_{CO₂} of 1.2 g L⁻¹ d⁻¹, X of 1.8 g L⁻¹, and μ of 1.7 d⁻¹, with R_{N/P}, C_{c,g}, and L_i at 11, 4.5 %, and 200 μE m⁻² s⁻¹, respectively. Notably, Ns demonstrated higher nutrient uptake rates and BY_p compared to Cv, with total production costs of \$0.629/L and respective required cost coverages of \$0.138 and \$0.375 for Ns and Cv. This study offers a comprehensive guideline for large-scale microalgae cultivation technology (MCT) processes, providing insights into optimizing conditions for biofuel production while considering economic viability.

1. Introduction

The use of microalgae for nutrient removal from wastewater (Ww) is an environmentally friendly and sustainable approach that capitalizes on the natural abilities of these microscopic organisms (Wicker and Bhatnagar, 2020; AlMomani and Örmeci, 2016; Almomani, 2019). This approach aligns with the principles of circular economy by turning waste into a valuable resource and contributes to the development of eco-friendly and cost-effective solutions for Ww management (Díaz et al., 2022; Gao et al., 2023; Qian et al., 2024; Skouteris et al., 2020). During the wastewater treatment (WwT) process, microalgae undergo photosynthesis process consuming nutrients, organic matters and CO₂ and producing oxygen. Oxygenating the Ww offers numerous benefits for the aquatic environment, including supporting aerobic bacteria that break down organic matter and helping to reduce foul odors (Skouteris et al., 2020; Almomani et al., 2022, 2023). Microalgae can also assimilate various emerging pollutants including heavy metals and toxins

contributing to improving water quality (Al Ketife et al., 2020; Almomani and Bhosale, 2021; Almomani et al., 2019a). In this context, algal cultivation process (ACP) is used to address both economic and environmental challenges (AlMomani and Örmeci, 2020).

The growth of microalgae in Ww depends mainly on water quality parameters such as pH, nutrients (nitrogen (N) and phosphorus (P)), availability of CO₂ and organic carbon (Kong et al., 2021; Almomani, 2020). An adequate supply of nutrients (N & P) is required to maintain algae growth and achieve optimal metabolic processes in algal cells. Microalgae utilize these elements as primary nutrients for cell growth (Gao et al., 2023; Qian et al., 2024). The primary mechanisms for phosphorus (P) and nitrogen (N) uptake by algae include assimilation and absorption (Carrillo et al., 2020). However, excessive amounts of nutrients can lead to extreme algae growth, resulting in algal blooms and water pollution. The availability of CO₂ is an important factor for the photosynthesis process and thus the growth of algae. Adequate levels of dissolved CO₂ in the water can promote algae growth. However, excessive CO₂ concentrations can lead to harmful effects, such as the

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Nomenclature		L_I	light intensity ($\mu\text{E m}^{-2} \text{s}^{-1}$)
$\%RE_{TN}$	Total nitrogen removal efficiency (%)	Ns	Nannochloropsis sp.
$\%RE_{TP}$	total phosphorus removal efficiency (%)	P	Biomass productivity ($\text{g L}^{-1} \text{d}^{-1}$)
μ	Specific algal growth rate (d^{-1})	P	Biomass productivity ($\text{g L}^{-1} \text{d}^{-1}$)
ACP	Algal Cultivation Process	PE	photosynthesis efficiency (%)
BBD	Box Behnkin Design	PV	probability value
B_Y	Biofuel yield (%)	R^2	Determination coefficient
BY_P	Biofuel productivity ($\text{g L}^{-1} \text{d}^{-1}$)	R_{CO_2}	CO_2 bio-fixation rate ($\text{g L}^{-1} \text{d}^{-1}$)
$C_{c,g}$	Input concentration of CO_2 (v/v%)	RMS	Root mean square
Cv	Chlorella vulgaris	RMSE	Root mean square of errors
DCW	Dry cell weight (g/L)	$R_{N/P}$	Nitrogen to phosphorus (P) ratio
DWw	Domestic wastewater	RSM	Response Surface Methodology
FES	Four Element Strategy	Ww	Wastewater
HRT	Hydraulic Retention Time (d).	WwT	Wastewater treatment
IC	Inorganic carbon	X	Biomass accumulation (g L^{-1})
ICMs	Inorganic carbon concentration mechanisms	Y_N	Nitrogen-algal biomass yield (%)
		Y_P	Phosphorous-algal biomass yield (%).

acidification of water (Almomani, 2020; Al Ketife et al., 2019). The acidity or alkalinity of the water (i.e. the pH) has a major effect on algal growth. Different algae species have different pH tolerances, and changes in pH can affect the availability of nutrients and the overall chemical balance of the water, thereby impacting algae growth (Choi et al., 2021).

Although water quality parameters are essential for algae growth, water movement, salinity, the availability of solar light are vital factors (Almomani and Örmeci, 2020). Light intensity (L_I), its duration, and quality are key factors affecting algal growth and photosynthetic process where the sunlight is converted into energy (Wang et al., 2008). Different types of algae have different light requirements, with some thriving in low-light conditions and others requiring intense sunlight (Li et al., 2021). Insufficient L_I can impede algal growth, while an excess L_I may result in photooxidative damage to algal cells. Different studies showed that microalgae can attain photosynthesis efficiency of 80 % under low L_I conditions (Høgslund et al., 2023; Mona et al., 2020; Fahrion et al., 2023; Chakravarty and Mallick, 2022). Algae also require specific temperature ranges within which they can grow optimally. Different species have different temperature preferences, but in general, warmer temperatures tend to promote faster growth (Li et al., 2021). Extreme temperatures, either too high or too low, can inhibit algae growth or even cause cell damage (Song et al., 2023; Aslam et al., 2021). The growth of algae, influenced by the flow and movement of water, varies depending on water speed; stagnant or slow-moving waters foster certain species, while faster currents suit others, impacting nutrient availability, light exposure, and algae cell dispersal (Almomani, 2020; Znad et al., 2019).

The ability of microalgae to uptake CO_2 gas as an essential element to produce biomass during the ACP has different practical implications. Algal bio-mitigation of CO_2 is considered a promising technique to overcome higher global warming problem (Almomani, 2019). For the last 50 years, researchers have investigated the potential method of using different algal strain for CO_2 bio-fixation (Farrelly et al., 2014; Yoo et al., 2010; Jacob-Lopes et al., 2009; Singh et al., 2016) and suitable use of produced algal biomass (Pulz and Gross, 2004a, 2004b). Microalgae can typically capture CO_2 from atmosphere and from industrial flue gases (Onyeaka et al., 2021; Abu-Khader, 2006). However, the biomass production may be limited by low CO_2 concentration in the air (Wang et al., 2008; Cuellar-Bermudez et al., 2015a). Against this, higher CO_2 concentration can promote the algal growth (Bilanovic et al., 2009). The CO_2 bio-fixation capability of a specific algae is positively related to an algal growth rate and light unitization efficiency (Fahrion et al., 2023; Chakravarty and Mallick, 2022; Jacob-Lopes et al., 2009; Li et al., 2022a; Cuellar-Bermudez et al., 2015b).

The use of cultivated algae biomass for biofuel production and the synthesis of other valuable chemicals is a promising and sustainable approach that has gained significant attention in recent years (Almomani et al., 2023; Ahamed et al., 2024). This method allows for the simultaneous production of biofuel and various high-value products (Høgslund et al., 2023; Li et al., 2022a, 2022b, 2020). With high lipid content, fast growth rates, and the ability to thrive in diverse environments, microalgae have several advantages over traditional biofuel feedstocks (Foroutan et al., 2018). Different studies were developed to extract the lipid content, particularly triglycerides from microalgae and convert is to biodiesel (Uma et al., 2023; Choo et al., 2020; Ebhodaghe et al., 2022). Besides biofuel, microalgae biomass can be used for the co-production of various value-added products including proteins, pigments, and carbohydrates, polymers expanding their economic viability and making them more resource-efficient (Ahamed et al., 2024; Levasseur et al., 2020; Almomani et al., 2022; Mukhtar et al., 2023; Hajilary et al., 2019).

While the potential of microalgae for biofuel and added-value products is significant, challenges such as cost-effectiveness, scalability, and technological advancements and processing methods need to be addressed. Design of experiments (DOE) using statistical methods is recognized as a highly effective and powerful approach for identifying key factors in multivariable systems to optimize various chemical and biological processes. Response Surface Methodology (RSM) coupled with sophisticated statistical tools is particularly useful for examining the impact of multiple factors and their interactions. The use of RSM and other statistical methods presents challenges, mainly due to their reliance on extensive experimental data. This poses difficulties in terms of the time required to complete experiments with high accuracy for successful application. Therefore, integrating biokinetic models with RSM is essential for addressing these issues and achieving more precise outcomes.

Research works that focused on the development of innovative solutions to optimize microalgae-based biofuel production are limited. Therefore, the objective of this work is to develop an approach that can be used to optimize biofuel production from microalgae biomass.

statistical method based on four elements growth parameters (algal strain, nutrient concentrations, energy inputs, and CO_2 concentrations) combined with bio-kinetic were used to investigate the influence of the parameters on the efficiency of the ACP, assessing the desirability of the responses factors and determining the optimal conditions for achieving maximum algal cultivation results. In addition, an economic study to assess the feasibility and viability of the proposed model were performed. The study involves comparing the growth, and the economic considerations for the production of biofuel from two microalgae strains

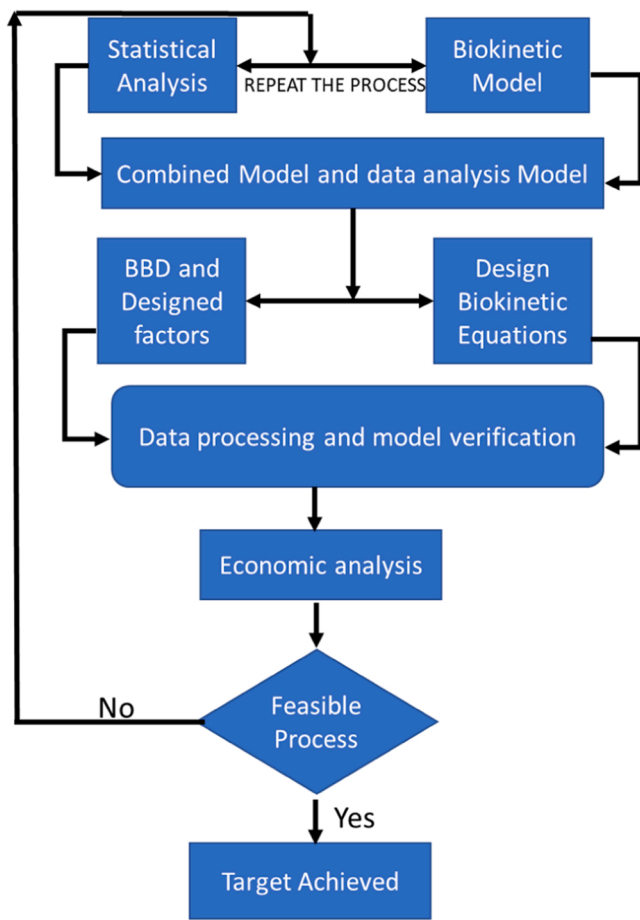


Fig. 1. Algorithm used in the study.

(freshwater strain - *Chlorella vulgaris* (Cv) and saline strain - *Nannochloropsis* sp. (Ns)). These strains were selected because they have high lipid content and they are the mostly used strains in Wwt.

2. Research methodology

2.1. Process development

Fig. 1 presents the flowchart used in optimizing the algal growth and biofuel production. The process begins with the development of a bio-kinetic model, which typically predict how living organisms behave under specific set of conditions using set of statistical data. The model includes set of mathematical equations that describe the growth rate of microalgae. The developed biokinetic model is then used to process data and verify growth rate against experimental values. This step ensures that the model accurately reflects observed data and can reliably predict outcomes. Box-Behnken Design (BBD) was combined with response surface methodology (RSM) to evaluate the effects of input parameters (L_b , $C_{c,g}$ and $R_{N/P}$) on the corresponding outcome parameters (R_{CO_2} , μ and X). The verified model and the data from the BBD were combined for further analysis. This involves statistical fine-tune of the model to improve its predictive capabilities. At this point, a thorough statistical analysis including (validation of the model, assessing the significance of variables, and determining the robustness of the process) is conducted. In case the results are not satisfactory, the process returns to the bio-kinetic model stage to adjust. This iterative approach allows for continuous improvement of the model based on feedback from the data analysis. Once the model and its equations are satisfactory, an economic analysis is performed to assess the cost-effectiveness of the process, balancing the operational costs against the benefits and potential

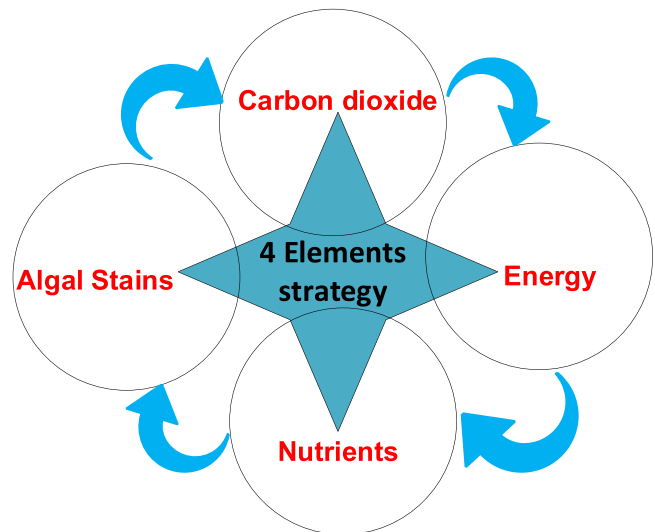


Fig. 2. Four elements strategy.

revenue from the product. A decision point is reached where the economic feasibility is acceptable. If the process is not economically feasible, iteration goes back to the statistical analysis and biokinetic for optimization. If the process is deemed economically feasible, the target of the project has been achieved. Statistical Software JMP from SAS was used in this analysis. It is worth to mention that the developed biokinetic models that determine the nutrients removal rate and biofuel production were developed using MATLAB software and were calibrated using real experimental values.

2.2. Optimization of biofuel production

Fig. 2 presents the four elements strategy (FES) used in optimization of biofuel production. The input to this strategy includes the key factors that influence the performance and cost of the algal process (i.e. concentration of CO_2 gas ($C_{c,g}$), energy input to the process, concentration of nutrients, and the type of algal strain). These inputs have direct influence on Algal Cultivation Process (ACP) in term of percent total nitrogen and phosphorous removals ($\%RE_{TN}$ $\%RE_{TP}$), biomass accumulation (X , g L^{-1}), and consequently the expected biofuel yield (B_y , %) and biofuel productivity (BY_p , g $L^{-1} d^{-1}$). Thus, their optimization is crucial in determining the overall efficiency and economic feasibility of the strategy. As the specific growth rate (μ , d^{-1}) is the main parameter that correlate the time required for specific algal cells to produce a certain amount of biomass, it was connected to biomass accumulation.

2.3. Bio-kinetic model

The specific algal growth rate (μ , d^{-1}) was determined as per Eq. (1). It was assumed during the development of this equation that the percentage of first cells inoculated to the ACP is 10 % of the final cell's concentration

$$\mu = \frac{\ln 10}{HRT} \quad (1)$$

Where; HRT is the hydraulic retention time (HRT, d). The biomass productivity (P , g $L^{-1} d^{-1}$) was correlated to the algal growth rate (μ , d^{-1}) and biomass accumulation (X , g L^{-1}) as per Eq. (2):

$$P = 0.39X\mu \quad (2)$$

The change in biomass accumulation as function of time was established in Eq. (3):

$$\frac{dX}{dt} = \mu X \quad (3)$$

The depletion in Nitrogen (N) and phosphorus (P) concentration during the cultivation period were established based on Eqs. (4) and (5):

$$\frac{dN_c}{dt} = -Y_N \frac{dx}{dt} \quad (4)$$

$$\frac{dP_c}{dt} = -Y_P \frac{dx}{dt} \quad (5)$$

Where; Y_N is Nitrogen algal biomass yield (%) and Y_P Phosphorous algal biomass yield (%).

2.4. Statistical model

The optimum output values were determined using quadratic model Eq. (6):

$$O_{out} = \alpha_0 + \sum_{i=1}^k \alpha_{ii} X_i^2 + \sum_{i=1}^{k-1} \sum_{j=1}^k \alpha_{ij} X_i X_j + \varepsilon \quad (6)$$

Where O_{out} is the response variable of the output parameter; i and j are patterns index number; α_0 is coefficient constant; α_i , α_{ii} and α_{ij} are the linear, quadratic, and interaction influence coefficients, respectively; x_i and x_j are the variables to be investigated calculated as per Eq. (7):

$$x_i = \frac{X_i - X_0}{\Delta X_i} \quad (7)$$

where X_i is the coded value of the independent variables, X_i is the actual value of the independent variable, X_0 is the actual value of the factor at

2.6. Biofuel production

The biofuel productivity (BY_P , g L⁻¹ d⁻¹) was estimated based on the obtained biomass accumulation (X , g L⁻¹) and specific growth rate (μ , d⁻¹) for two selected algal strains following Eq. (9) (Znad et al., 2019):

$$BY_P = X \mu B_Y \quad (9)$$

In this equation the biofuel yield (B_Y , %) was calculated from lipids density and algal biomass according to Eq. (10) (Zhu et al., 2013):

$$\%B_Y = \frac{\text{Weight of Extracted Lipid}}{\text{Weight of Biomass}} * 100\% \quad (10)$$

2.7. Economic analysis

The cost analysis of the conversion of biomass to biofuel was carried out in different steps; (1) determine baseline cost of cultivation of 1 m³ of algal media at HRT = 2 days. The baseline costs were associated to the four elements ($C_{c,g}$ %, algal strain, energy and nutrients), (2) calculate the specific costs for CO₂ gas bio-fixation and the Hydrothermal Liquefaction (HTL) process for each strain (N_s and C_v), (3) analyze output costs for each strain such as the cost of required nutrients per kilogram of biomass and the cost for nutrients uptake and (4) determine algae production and oil extraction costs, including the cost of delivery of the wastewater and CO₂ to the Algal photo-bioreactor (PBR), mitigated by the value of the recovered resources such as the biofuel, residual organic carbon (e.g, biogas), and nutrients.

Based on the above points, the overall cost of production of one litter (1 L) of biocrude (B_{CO}) was determined using Eq. (11):

$$B_{co} (\$/L) = (C_{Cr} + RN_{Cr} + RP_{Cr} + Fg_{Cr} + Ae_{Cr} - In_{CO} - Dfg_{CO} - Cf_{CO} - Ab_{CO} - Lab_{CO} - Dis_{CO} - WW_{CO} - Sd_{CO} - HTL_{CO} - Solar_{CO} - Pre_{CO}) \quad (11)$$

the centre point, and ΔX_i is the step change value. The multi-level experimental design, include both minimum and maximum levels of operational ranges of $R_{N/P}$, $C_{c,g}$, and L_f .

Experimental results obtained from Fifteen experimental runs were used to determine the coefficients of the polynomial as per Eq. (6). This number of experiments were determine based on input variables to be investigated and the of levels each factor with respect to optimized parameter.

2.5. Response parameters and desirable function

Desirability functions was used to optimize the APC process based on multiple responses. The idea is to transform each response into a scale that ranges from zero (completely undesirable) to one (fully desirable). This scale facilitates the simultaneous optimization of multiple responses by converting them into a single composite desirability score. Therefore, this approach is beneficial to recognize conflicting responses and to find the best overall set of conditions. In this study, three-dimension scatter plot combined the desirability factor was used to determine the most desirable combination of responses, Eq. (8):

$$D = (D_1(R_1)D_2(R_2).....D_i(R_i))^{1/k} \quad (8)$$

Where R_1 , R_2 and R_i represent the response of any of the key parameters and $D_1(R_1)$, $D_2(R_2)$ and $D_i(R_i)$ are the desirability of key parameters, k denoting the number of responses. In this equation the overall desirability, D , is computed by multiplying the individual desirability for each response, all of which are scaled from 0 to 1 (Zhu et al., 2013).

where B_{CO} is the biocrude production cost (\$/L), the items C_{Cr} , RN_{Cr} and RP_{Cr} refer to the credit from carbon, nitrogen and phosphorous removed from the wastewater, respectively, Fg_{Cr} and Ae_{Cr} are the flue and aeration gas delivery costs. In_{CO} is the cost of inoculation, Dfg_{CO} is the cost of delivered flue gas, Cf_{CO} is the cost of cultivation, Lab_{CO} is the cost of labour, Ab_{CO} is the cost of abiotic WW treatment, Dis_{CO} is the cost of disinfection, WW_{CO} is the cost of delivered WW , Sd_{CO} is the cost of solar drying, and HTL_{CO} is the cost of biocrude production. All costs were reported in \$/L. The revenue achieved from the conversion of the algal biomass was calculated based on Eq. (12). The cost items in Eq. (12) is based on operating cost of the process (OPEX) that includes energy, materials, land, maintenance, insurance, loan payments, taxes, and labour.

$$Rev_{ue_{biocrude}} = \sum cost_{out} - \sum cost_{in} \quad (12)$$

The costs associated with extraction of biomass lipids and the conversion to biocrude via HTL as well as the upgrading processes was correlated to the price for the algal biomass with a given oil content (Zhu et al., 2013). This item was determined based on the ability to produce equivalent amount of energy (MJ) as compared to crude oil following on (Eq. 13):

$$m_{algal} = \frac{E_{petroleum}}{f_{xoil} E_{biocrude}} \quad (13)$$

where m_{Algal} is the quantity of algal biomass (kg) with the energetic equivalent of a litre of crude petroleum, $E_{petroleum}$ is ~45 MJ/kg_{crude oil} at a density of 850 kg m⁻³ and $E_{biocrude}$ is the energy content of the algal oil in MJ/kg. The latter has been reported as being 38 MJ kg⁻¹ at an assumed density of 887 kg m⁻³. Assuming the conversion of 1 L of crude

Table 1

Algae strains growth, biomass productivity, CO₂ and nutrients requirements, % nutrient removals and biofuel production as a function of media composition.

Cult. Media	Algal Strain	Sys. conf.	IOc, % wt	C, cg, % v/v	Qg, vvm	L _i μE m ⁻² s ⁻¹	T, °C	pH _i	Cd, d	TN _i , mg L ⁻¹	TP _i , mg L ⁻¹	R _{N/P}	μ, d ⁻¹	% RE _{TP}	% RE _{TN}	X _i , g L ⁻¹	P _{X_i} , mg L ⁻¹ d ⁻¹	B _Y , % DCW	B _{YP} , g L ⁻¹ d ⁻¹	Ref
MLA	Cv	C _{PBR}	10	0.03 ^a	nr	180	25	7.5	10	0–56	0–19	7–10	1.04	100	100	1.58	nr	7	0.11	(Alketife et al., 2017)
D _{WW}	Cv	OP _{PBR}	11	0.03	0.02	270–310	25	7.2	15	nr	nr	1–10 11–20 21–30 61–70	nr	89 81 59 24	78 84 83 73	nr	2.75 2.3 1.18 0.41	50 ^b	nr	(Tango et al., 2023)
AM 40 % C ^c	<i>N.gaditana</i>	B-C ^c	10	0.03	0.2 ^d	500	20	7.8–7.9	6	109	22.4	4.9	nr	58	20–40	2.34	0.39	20–25	nr	(Sepúlveda et al., 2015)
50 % C						500				149.1	4.75	31.3	nr	80–90	20–40	2.34	0.39			
AM 40 % C						300				260.3	8.18	31.8	nr	80–90	20–40	2.4	0.4			
50 % C						300				109	22.4	4.9	nr	42	40	1.75	0.29			
						300				149.1	4.75	31.3	nr	80–90	25	1.75	0.29			
						300				260.3	8.18	31.8	nr	80–90	39	1.8	0.3			
Syn Dom.	Cv	PBR _s	nr	0.04	nr	22	15–22	nr	10	7.48–22.1	1.69–2.2	9.7–12.9	0.59	>95	>95	0.2	0.033	50 ^b	0.06	(Marbelia et al., 2014)
Mww	Cv	MPBR	nr	0.04	nr	nr	nr	nr	8	7.48–22.1	1.7–2.17	15.5–22	0.17			0.2–0.75	0.06	19	nr	(Yang et al., 2016)
	<i>S. obliquus</i>	VFPPBR	nr	0.04	nr	nr	nr	nr	8	12	1.8	6.6	1.39	>99	>99	nr	nr	29	nr	
P _{WW}	Cv	EF _{PBR}	nr	0.04	nr	30.2	20	7.9	30	40	10	4.1	nr	35.2	60.2	2.71	0.09	50 ^b	nr	(AlMomani and Örmeci, 2016)
S _{WW}								7.5		44	26	2.42	nr	11.9	55.9	1.86	0.06		nr	
C _{WW}								4.9		130	200	0.68	0.25	25.9	33.6	2.37	0.08		0.6	
P _{WW}	<i>N. oleoabundans</i>												0.35	32.4	50.7	0.5	nr		nr	
S _{WW}													0.26	5.67	56.8	1.44	nr		nr	
C _{WW}													0.08	19.8	61.6	1.11	nr		nr	
P _{WW}	MIM												0.23	70	63.2	0.58	nr		nr	
S _{WW}													0.21	30.8	67.3	0.62	nr		nr	
C _{WW}													0.41	50	80.8	1.03	nr		nr	
PLA	<i>A. platensis</i>	GBC _{PBR}	nr	0.04	2	30	30	7	8–10	75	5.32	14	nr	75	100	2	nr	6	nr	(Markou et al., 2015)
	Cv ^f													40	100	1.3	0.11	22	nr	
MLA	Cv ^g	C _{PBR}	10	0.04	1	185	25	6.9	13	28	6	4.7	0.62	79	100	1.02	nr	11	0.07	(Znad et al., 2018)
P _{WW}								9.3		30.6	6.6	4.6	1.2	80	100	1.62	nr		0.21	
S _{WW}								9		11.8	5.6	2.1	0.66	83	100	1.16	nr		0.08	
PE								9.3		23	1.9	12.1	0.31	78	100	0.69	nr		0.02	
P _{WW}	<i>D. communis</i>	PyPBR	5	0.04	nr	100	30–27	8.3	7–14	110	36	3	nr	>50	>99.9	0.38	0.038	50 ^b	nr	(Lavrinovičs et al., 2020)
	<i>T. obliquus</i>															0.21	0.021			
	<i>C. protothecoides</i>															0.4	0.049			
S _{WW}	<i>D. communis</i>							8.2		40	36	1.1				0.36	0.02			
	<i>T. obliquus</i>															0.36	0.032			
	<i>C. protothecoides</i>															0.35	0.021			

Table 2
CO₂ biofixation (R_{CO2}, P_x and R_x) under different cultivation conditions and algal strains.

Mode of Operation - and C. medium	Algal Strain	TN _{in} , mg L ⁻¹	TP _{in} , mg L ⁻¹	C _{c,g} , % v/v	L _i , μE m ⁻² s ⁻¹	T, °C	pH _i , Value	HRT, d ⁻¹	μ, d ⁻¹	X _{max} , g L ⁻¹	R _{CO2} , g L ⁻¹ d ⁻¹	R _x , g L ⁻¹ d ⁻¹	% RE _{TN}	% RE _{RE}	B _y , %	B _{YP} g L ⁻¹ d ⁻¹	Refs
LS _{PBRB} -SY _{Ww}	<i>S. platensis</i>	nr	nr	6	84.2	27	nr	nr	0.26	nr	1.03	0.68	nr	nr	5	nr	(Singh et al., 2016)
EF _{PBRB} -MZM		618	88.9	2.5	400*	28–30	9.4	160	nr	8.75	0.855	1.5	nr	nr	50 ^b	nr	(Ho et al., 2018)
EF _{PBRB} -TMSYTF _{Ww}		130	15	nr	90	25	7.79	17	nr	0.76	0.042	0.077	93.3	79.96	50 ^b	nr	(Zhou et al., 2017)
CP _{BRC} -ZM**		444	82.5	5	180±5	30.0 ± 0.5	nr	35–50	nr	4.4	0.366	0.67	nr	75.21	50 ^b	nr	(Liu et al., 2018)
CP _{BRC} -ZM					5	0.5	nr	nr	nr	3.53	0.293	0.54	nr	67.34			
AA-SDCG-BC, -CF		76	89	100	nr	nr	nr	8–9	0.41	1.83	0.417	0.787	70	19	50 ^b	0.37	(Kumari et al., 2014)
PS _{PBR} -S _{Ww}		53.3	9.3	2.5	220	23	7.1	nr	1.48	0.68	0.22	0.42	87	84	50 ^b	0.5	(Almomani et al., 2019b)
				5					1.6	0.76	0.25	0.44	88	86		0.6	
				10					1.7	0.8	0.31	0.54	93	90		0.68	
				15					1.4	0.65	0.26	0.43	88	86		0.45	
				20					1	0.68	0.19	0.29	76	74		0.34	
				35					0.82	0.47	0.15	0.24	87	84		0.2	
CMP _{BRC} -Aq _{Ww}	<i>C.v</i>	6.81	0.42	0.07	121	nr	nr	1	0.17	0.04	0.04	0.07	82.7	86.1	50 ^b	0.003	(Gao et al., 2016a)
CMP _{BRC} -D _{Ww}		7.5–22.1	1.7–2.17	0.1				2–5	0.17	0.2–0.75	0.06	0.17	nr	nr		0.01–0.06	
MPBR-S _{Ww}		15	0.8	4	128	25	nr	2	nr	0.95	0.049	0.092	64 ± 6	85 ± 3	50 ^b	nr	(Gao et al., 2015)
BMPBRC-S _{Ww}									nr	1.37	0.072	0.135	83 ± 4	86 ± 2		nr	
SF-MBM		41.2	53.1	0.7–2.1	20	25	nr	10–50	nr	1.8	0.18	0.31	nr	nr		nr	Ben Amor-Ben Ayed et al., (2015)
																	Ben Amor-Ben Ayed et al., (2015)
MPBRC-Dse		0.78	14.12	4	121	25	6.5–7.8	2	nr	1.73	0.05	0.094	87.7	76.6	50 ^b	nr	(Gao et al., 2016b)
BMPBR-SE		15	0.8	4	142	25–28	6.8–7.5	2	nr	1.37	0.072	0.12	nr	85	50 ^b	nr	(Gao et al., 2015)
C _{PBR} -MLA		28	7	0.04	200	24	7.5	50	0.64	1.02	0.123	0.19	68	100	50 ^b	0.32	(Al Ketife et al., 2017)
				0.5			7.5		0.77	1.67	0.18	0.33	90	100		0.64	
				2			7.3		nr	1.89	0.2	0.41	92	100		nr	
				5			6.1		nr	2.93	0.324	0.61	100	100		nr	
				22			5.8		0.09	1.12	0.12	0.26	60	60		0.05	
RW _{PBR} -CFM	<i>A. sp.</i>	nr	nr	10.6	460	nr	8	3.3	nr	0.76	nr	0.43	nr	nr	50 ^b	nr	(de Godos et al., 2014)
CV _{PBR} -BGATCM	<i>S. obliquus</i>	nr	nr	0.03–35	107	25	nr	0.76–0.19	0.64	1.39	nr	0.097	nr	nr	20.8–41.7	0.44	(Basu et al., 2014)
PS _{PBR} -S _{Ww}	<i>MIMA</i>	53.3	9.3	2.5	220	23	7.2	nr	nr	0.81	0.4	0.79	90	88	50 ^b	0.71	(Almomani et al., 2019b)
				5					1.2	0.9	0.45	0.87	93	91		0.54	
				10					nr	0.95	0.56	1.1	99	96		0.99	
				15					1.71	0.79	0.47	0.81	100	98		0.67	
				20					1.24	0.81	0.34	0.58	85	83		0.5	
PS _{PBRB} -F/2	<i>N.S</i> ^c	nr	nr	3	100	30	7–8	0.4–0.8	0.09–0.18	0.4–0.8	0.05–0.04	0.72–1.44	nr	nr	50 ^b	0.01	(Raees, 2014)
CB _{PBRB} -F/2	<i>N.S</i>	30	6	99.9	257 ± 11	30	7	2.1	0.3	2.1	nr	0.54	96	100	50 ^b	0.31	(Kim and Lee, 2016)

^B : for batch; _s=semi- continues cultivation process PBR = Photobioreactor; LS_{PBR} laboratory scale PBR; SY_{Ww} synthetic wastewater; ^b Algal biofuel fraction taken 50 wt% as previously reported (Al Ketife et al., 2019); EF_{PBRB} erlenmeyer flask PBR; MZM modified Zarrouk medium; TMSYTF_{Ww} treat mixed synthetic toilet flushing wastewater of seawater; CP_{BRC} cylindrical or column PBR; ZM Zarrouk medium; ** continuous cultivation process for five months; AA-SDCG-BC air agitated - Sintered disk chromatographic glass bubble column- Bubble column; CF complex fertilizer; PS_{PBR} pilot scale PBR; CM_{PBR} cylindrical membrane PBR; Aq_{Ww} aquaculture wastewater; MPBR membrane PBR; BMPBRC biofilm membrane PBR; SF Shake flask; DSE domestic secondary effluent; SE secondary Effluent; RW Race way PBR; CFM commercial fertilizer medium; CV culture vessel PBR; MIMA mixed indigenous microalgae; PS parallel system PBR; ^c Nannochloris. sp. CB cylindrical bottle PBR; A sp. *Anabaena* sp.

oil to various useable transport energy products costs roughly the same as converting X kg of biomass to bioenergy, the maximum acceptable price that could be paid for the biomass would be the same price of a barrel of crude petroleum. This parameter can be used as the break-even selling price (BESP) Eq. (14):

$$\text{BESP} \left(\frac{\$}{\text{L}} \right) = \frac{\text{Price of 1 barrel of petroleum}}{\text{No. of equivalent litre of biocrude}} \quad (14)$$

All the determined costs were then compared between the two algal strains and used to evaluate the feasibility of each.

3. Results and discussion

3.1. Nitrogen and Phosphorus removal mechanism

Table 1 summarizes the most common studies that focused on nutrient removal by algae strains. A thorough examination of the conditions under which algae are cultivated reveals a range of conditions concerning various strains, culture mediums, and system setups. Notably, *Cv* grown in a cylindrical column photo bioreactor (CCPBR) that uses microbiology-derived medium (MDM) with TN_i and TP_i concentrations in the range $0\text{--}56 \text{ mg.L}^{-1}$ and $0\text{--}19 \text{ mg.L}^{-1}$ demonstrated an excellent $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}}$ of 100 %. The same algal strain grown in domestic wastewater (DWw) exhibited $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}}$ in an optical panel photo bioreactor (OPPBR) in the ranges 78–73 % and 24–89 %, respectively. The $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}}$ of *N. Gaditana* cultivated in an artificial medium (AM) bubble Column (B-Cc) system, ranged from 20 % to 40 %, and 58–90 %, respectively. Several PBRs that contain synthetic medium and *Cv* demonstrated high $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}}$ of >95 %. *Cv* cultivated in a vertical Flat plate PBR (VFPPBR) showed $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}} \geq 99$ % from municipal wastewater (Mww). Tests performed in Erlenmeyer flask PBR (EFPBR) using *N. Oleoabundans* treating primary Wastewater (PWw) achieved $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}}$ of 60.2 % and 35.2, respectively. *A. Platensis* were able to achieve $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}}$ of 100 % and 75 % in a glass bubble column photo bioreactor (GBCPBR). Moreover *D. Communis*, *T. Obliquus* and *C. Protothecoides* grown in Pyrex bottle PBR (PyPBR) showed $\% \text{RE}_{\text{TN}}$ and $\% \text{RE}_{\text{TP}}$ of ≥ 99.9 % and 50 % from secondary wastewater (SWw). All the previous results highlight the versatility and possible uses of different algal strains for nitrogen removal under varied growing circumstances. The relation between biofuel production and media composition shows that saline algal strain, *NS* have more capacity to remove N and P compared with freshwater algal strain *Cv*, and consequently higher biomass and biofuel production is expected. Furthermore, slain strain can stand the harsh weather conditions compare with the fresh ones (see Table 1).

Sys. Conf system configuration; Ioc Initial inoculum; C_{cg} inlet CO_2 gas concentration; Q_{g} inlet gas-flowrate; L light intensity; C_{d} Cultivation days; TN_i Total inorganic nitrogen concentration; TP_i Total inorganic phosphorus concentration; N/P Nitrogen to Phosphorus ratio; μ Specific growth rate; TP , TN % RE removal efficiency; P_x biomass productivity; B_y Biofuel Yield; MLA Marine labs American society of microbiology-derived medium; ^a Atmospheric level; nr not reported; PBR photo bioreactor; C_{PBR} Cylindrical column PBR D_{Ww} Domestic Wastewater; OP V-Ccut Optical Panel PBR; ^b Algal biofuel fraction taken 50 wt% as previously reported (Al Ketife et al., 2019); B Batch process; C continuous process; ^c 0.25 % of algal culture was harvested and replaced with fresh media; ^d pure CO_2 gas injected on demand 0.01 vvm; ^e centrate was mixed with algal medium at 40:60 mixing ratio; Syn Synthetic; Dom Domestic; MPBR membrane PBR; Mww Municipal wastewater; Ww wastewater; P_{Ww} Primary Ww; S_{Ww} Secondary Ww; C_{Ww} Centrate Ww; EFPBR Erlenmeyer flask PBR; MIM Mixed indigenous Microalgae; PLA Poultry litter Ash; GBCPBR glass bubble column PBR; ^f algal stain verain is SAG 211–11b; ^g stain ID CCAP 211/11B, CS-42; PE petroleum effluent; Pyrex bottle PBR;

Microalgae cells have the ability to assimilate nutrients (N & P) from

wastewater, at the same time, generate a significant amount of sustainable biomass. Nitrogen is regarded as a crucial ingredient for the formation of microalgae cells. It can exist in the form of nitrate (NO_3^-), (NH_3), and urea ($\text{CH}_4\text{N}_2\text{O}$). These substances, individual or combined, are the most common sources of nitrogen in nature (Zarrinmehr et al., 2020; Lourenço et al., 2002). Algal cells may assimilate organic nitrogen, which originates from inorganic ammonium (NH_4^+), ammonia, and nitrogen gas (N_2). Ammonium is preferred as the desirable form of nitrogen since its digestion does not involve a redox reaction, resulting in lower energy requirements. Research has indicated that algae exhibit a preference for ammonium over nitrate, and the consumption of nitrate does not commence until the ammonium is nearly depleted (Fan et al., 2020; Salbitani and Carfagna, 2021).

Phosphorus (P) is also an element that is important to many biological components and has a major role in the metabolism of microalgae. It is an essential component of proteins, lipids, DNA, RNA, ATP (adenosine triphosphate), and components found in cell membranes. Its fundamental significance emphasizes how important it is to the general development and operation of microalgae. Photophosphorylation is one important phosphorus-related metabolic activity. Through assimilation by microalgae cells, inorganic phosphates, preferably in the forms of $\text{H}_2\text{PO}_4^{1-}$ & HPO_4^{2-} , are transformed into organic forms. This metabolic process is a component of the energy-generation process, which turns energy intake into adenosine diphosphate (ADP) and ATP (Martínez et al., 1999; Lavrinovičs et al., 2021; Yaakob et al., 2021; Wu et al., 2021). It is important to remember that absorption into the cell is not the only factor controlling phosphorus removal in wastewater. Phosphorus dynamics are also significantly influenced by external factors like pH and dissolved oxygen levels. Since phosphorus is a non-gaseous element, under some circumstances it tends to precipitate out of the medium. Phosphorus can precipitate specifically in response to high dissolved oxygen concentration and pH increases. This phenomenon impacts the total effectiveness of phosphorus removal; hence it is critical to consider in WwT operations. Furthermore, phosphorus elimination may also take place as a result of adoption process on the algal cells. It is possible that microalgae can remove phosphorus from the surrounding environment in addition to using it for their metabolic functions (Robles et al., 2020). Understanding the complex interactions among phosphorus dynamics, environmental factors, and microalgae metabolism is essential for enhancing WwT approaches and guaranteeing sustainable methods in diverse applications, like nutrient recycling and water remediation (Al Ketife et al., 2020).

Principally, molar ratio of the water supply must be calculated in accordance with stoichiometric ratio of the algae biomass. The $R_{\text{N/P}}$ directly affects not only microalgae growth as they are a simulated simultaneously in algal cultivation process (ACP), but also the removal capacities of N and P in advanced WwT (Wang et al., 2010), therefore $R_{\text{N/P}}$ in cultivation medium should be in a proper range to satisfy a maximum N and P removal efficiently. $R_{\text{N/P}}$ ratio is usually presumed as 16:1. This ratio is not a common biochemical ideal, but instead represents an average of species-specific $R_{\text{N/P}}$ that vary from 8 to 45 (Alketife et al., 2017; Klausmeier et al., 2004a). Algal cells CO_2 -biofixation mechanism

Table 2 shows that the performance of various algal strains in term of CO_2 biofixation rates (R_{CO_2} , $\text{g.L}^{-1}.\text{d}^{-1}$, P_x , $\text{g.L}^{-1}.\text{d}^{-1}$ and R_x , $\text{gCO}_2.\text{g}^{-1}_{\text{biomass}}.\text{d}^{-1}$) using different strains and under different modes of operation: TN_{in} (mg.L^{-1}), TP_{in} (mg.L^{-1}), temperature, HRT, $C_{\text{c,g}}$ (%v/v) and L_i ($\mu\text{E.m}^{-2}.\text{s}^{-1}$). *S. Platensis* can resist $C_{\text{c,g}}$ in the range 2.5–35 %, L_i in the range $84.2\text{--}2020 \mu\text{E.m}^{-2}.\text{s}^{-1}$ and achieve P_x , R_{CO_2} and R_x in the ranges $0.042\text{--}0.855$, $\text{g.L}^{-1}.\text{d}^{-1}$, $0.07\text{--}0.78 \text{ g.L}^{-1}.\text{d}^{-1}$ and $0.15\text{--}0.68 \text{ gCO}_2.\text{g}^{-1}_{\text{biomass}}.\text{d}^{-1}$, respectively. The table underscores the significance of optimizing these parameters in algal cultivation, providing valuable insights into optimizing biofixation processes across different algal strains and environmental conditions. The overall trend shows that the growth of fresh and slain water strains increase by increasing the CO_2

Table 3
The effect of light intensity on algal growth parameters.

Cult. Med.	Algal strain	Sys. Confi.	TP _{inv} mg L ⁻¹	TN _{inv} mg L ⁻¹	C _{c,g} % v/v	L _B μE m ⁻² s ⁻¹	pH _i –	Culti. days	X _{max} g L ⁻¹	μ _i d ⁻¹	% RE _{TP}	% RE _{TN}	R _{CO2} g L ⁻¹ d ⁻¹	Lipids pr., P _{lipids} mg L ⁻¹ d ⁻¹	B _y , % wt	B _{yp} , g L ⁻¹ d ⁻¹	Ref.
MW	NS	M.B	4	36	0.03 ^a	50			0.47 0.36 0.27 0.12	nr nr nr nr	57 45 37 35	72 42 38 16	0.05 ¹ 0.04 ¹ 0.03 ¹ 0.01 ¹	0.008 ² 0.007 ² 0.003 ² 0.002 ²	50	nr	(Kang et al., 2015)
Z8	NS	nr	nr	nr	0.03 ^a	100			0.81 0.63 1.59 1.26 1.33	nr	nr	nr	0.20 0.16 0.40 0.31 0.33	0.013 ² 0.009 ² 0.031 ² 0.021 ² 0.026 ²	50	nr	(Hultberg et al., 2014)
MJ	NS	EF	nr	nr	0.03 ^a	100			0.78 0.84 0.8	nr nr nr	nr nr nr	nr	0.27** 0.29** 0.28**	nr nr nr	50	nr	(Kim et al., 2014)
AM	NS ^a	CB _{PBRB}	233	30	0.03	200	8	12	2.2	nr	100	99	3.96	0.16	21.79		(Jazzar et al., 2015a)
	C. sp. ^b	CB _{PBRB}				950			1.6	nr	96	93	3	0.26	45.62		
		CB _{PBRB}				200			1.77		99.79	99.79	0.17	nr			
		CB _{PBRB}				950			1.3		62.7	92.1	0.64	nr			
TM	C. vilgaris ^c	F _{PBR}	nr	nr	0.03	36	7	12	0.78	0.68	29.3	97.1	0.27	nr	50	0.26	(Gonçalves et al., 2014)
						60			0.92	0.83	35	97.2	0.32			0.32	
						120			0.99	1.07	42.7	94.6	0.36			0.53	
						180			nr	1.19	67.6	100	0.471			nr	
	P. subcapitata					36			0.42	0.64	34.2	88	0.14			0.13	
						60			0.62	0.94	34.2	98.5	0.21			0.3	
						120			0.64	1.09	51.3	93.3	0.23			0.35	
	S. salain ^d					180			0.798	1.09	51.3	100	0.27			0.43	
						36			0.69	0.59	35.4	92.5	0.21			0.2	
						60			1.07	0.89	34.7	98.4	0.32			0.47	
						120			0.93	1.09	37.9	95.5	0.27			0.5	
						180			1.259	1.09	36.6	99.1	0.39			0.7	
	M. aeruginosa ^f					36			0.73	0.54	29.7	97.3	0.24			0.2	
						60			1.07	0.78	36.4	99.1	0.35			0.4	
						120			0.99	1	36.2	95.2	0.32			0.5	
						180			1.174	1	41.1	100	0.39			0.58	
Ww	C. vilgaris	PC _{PBRB}	2.23	4	0.03	146	nr	nr	0.13	nr	100	100	0.24	nr	50	nr	(Shriwastav and Bose, 2015)
						14			0.1	nr	55	100	0.18	nr			
BG11	C. oleofaciens ⁱ	EF _{PBR}	nr	nr	0.03	50	7.5	20	2.5	0.14	nr	nr	4.7	70.25	28.16	0.09	(Rayati et al., 2020)
						100			3.9	0.24			7.02	92.87	31.75	0.3	
						200			7.35	nr			13.81	126.72	43.29	1.25	
						400			7	nr			13.16	198.32	59.18	1.55	
						800			0.55	0.01			1.03	50.36	26.35	0.001	
MLA	C. vilgaris	CPBR	7	28	0.03	100	7	10	1.12	0.72	100	80	0.37	nr	50	0.4	
						250			1.39	0.64	76	74	0.35			0.44	
						400			0.53	0.42	50	60	0.11			0.11	

MWw municipal wastewater; BM Basic medium; Z8 Zarrouk cultivation media; nr Not reported; EF_{PBR} Erlenmeyer flask PBR; AM algal medium.

^a Strain ID CCAP 211/11B; CB class bubble PBR

^b C. sp. Chlorella sp. ID is ID CCAP 278/4; TM Test Medium

^c strain ID LEGE06079; F_{PBR} Flask PBR

^f Strain ID LEGE 91344

ⁱ Stain ID is KF584224.1.; Ww wastewater; PC plastic container PBR; BG11 Blue green medium;

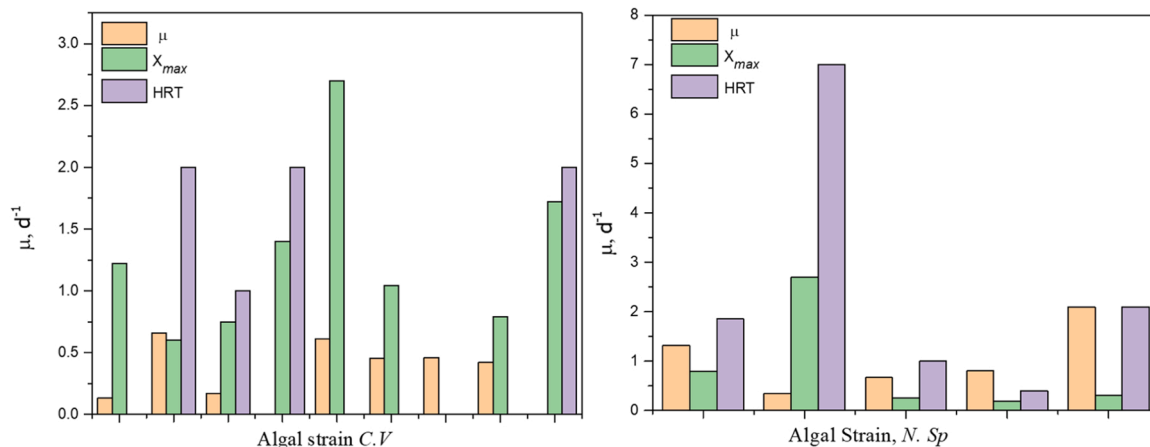


Fig. 3. Specific algal growth rate (μ , d^{-1}), maximum biomass accumulation ($X_{\max}$, g L^{-1}) and Hydraulic Retention Time (HRT, d) for the two algal strains; Cv (AlMomani and Örmeci, 2016; Gao et al., 2016b; Klausmeier et al., 2004b) and NS (Li et al., 2014; Takagi et al., 2000; Wahidin et al., 2013; Kim and Lee, 2016; Liu et al., 2013; Watanabe and Fujii, 2016; Teo et al., 2014; Jiménez-Pérez et al., 2004; Wileman et al., 2012) the most important parameters.

gas concentrations, $C_{c,g}$ from 0.03 % up to 10 %, whereas beyond this concentration algal growth was dropped down accordingly. Slain strains show higher CO_2 biofixation compared with freshwater strain (see Table 2).

When carbon dioxide (CO_2) dissolves in water, it forms an inorganic carbon (IC), a weak acid/base buffer system that includes carbonate (CO_3^{2-}), bicarbonate (HCO_3^-), and carboxylic acid (H_2CO_3). The most important buffer system found in natural water is the bicarbonate-carbonate system (Boyd, 2020; Middelburg et al., 2020; Marcandalli et al., 2022). IC in water can be assimilated by microalgae as substrates, which reduces CO_2 via IC concentration mechanisms (ICMs) (Burlacot et al., 2022; Iñiguez et al., 2020). After being used by microalgae cells, the carbon diffuses passively across the cytoplasmic membrane. Following this, carrier proteins actively transfer the material through the chloroplast envelope and plasma membrane. As indicated by reaction (15), light and water are required as sources of energy and electrons to finish the formation of sugars as a result of photosynthesis (Goria et al., 2024):



Light caused oxidation of the water, resulting in the simultaneous production of oxygen (O_2) along with the generation of reduced adenine dinucleotide phosphate (NADPH), adenosine triphosphate (ATP), and nicotinamide. The ICMs utilize active intracellular transport of energy to enhance the concentration of CO_2 at the location of the key enzyme, ribulose 1,5-bisphosphate carboxylase/oxygenase (Rubisco), which is responsible for carbon assimilation in photosynthesis. This mechanism allows for efficient CO_2 fixation rates, even at lower intracellular concentrations, within the Calvin Cycle. ATP and NADPH must be utilized prior to the implementation of carbon fixation in the Calvin Cycle. Interchange of different carbon forms takes place in the periplasmic space and is facilitated by the enzyme carbonic anhydrase (CA) (Bi et al., 2021; Rudenko et al., 2021). This enzyme supplies CO_2 to Rubisco by converting accumulated bicarbonate into CO_2 by dehydration (Bharti et al., 2021). The maximization of CO_2 absorption rate is dependent on various parameters, including as the specific algal strain, pH of the medium, availability of light, temperature, and concentration of CO_2 gas.

3.2. Light intensity and cells growth

The correlation between CO_2 fixation and L_I does not demonstrate any consistent pattern across various species or in different studies conducted on the same species (refer to Table 3). However, in individual investigations carried out under standardized conditions, both fixation

and biomass productivity exhibit an increase with L_I , as expected, reaching a maximum determined by light saturation (Pires et al., 2014; Shah et al., 2023; Esteves et al., 2020). Batch experiments involving four distinct algae species (Cv, *P. Subcapitata*, *S. Salina*, and *M. Aeruginosa*) have indicated that a threefold elevation in L_I from $36 \mu\text{mol m}^{-2} \text{s}^{-1}$ leads to a 70–90 % increase in biomass production and a 35–45 % increase in CO_2 uptake. Subsequent increments in L_I could potentially impede and reduce R_{CO_2} and μ (Acién Fernández et al., 2012). Increased L_I from 50 to $100 \mu\text{E m}^{-2} \text{s}^{-1}$ has significantly increased the BY_p from $0.002 \text{ mg L}^{-1} \text{d}^{-1}$ to $0.026 \text{ mg L}^{-1} \text{d}^{-1}$ following the same trend for R_{CO_2} of 96.9 % increasing recorded for microalgae strain NS. Literature data show the same trend for the Cv strain, as the R_C was increased from $0.27 \text{ g L}^{-1} \text{d}^{-1}$ to $0.47 \text{ g L}^{-1} \text{d}^{-1}$ when L_I increased from 36 to $180 \mu\text{E m}^{-2} \text{s}^{-1}$. Against this, BY_p dropped from $126.72 \text{ mg L}^{-1} \text{d}^{-1}$ to $50.6 \text{ mg L}^{-1} \text{d}^{-1}$ when L_I was increased from $200 \mu\text{mol m}^{-2} \text{s}^{-1}$ to $800 \mu\text{E m}^{-2} \text{s}^{-1}$, respectively (Table 3).

Light intensity (L_I) is a crucial element in the photosynthesis process and the growth of algae. The availability of L_I during ACP is determined by the concentration of cells, which is inversely related to the distance that light can penetrate into the algal culture. The L_I diminishes fast as a result of the shadowing caused by the algae cells (Chen et al., 2008). Nevertheless, by using a short light path length and ensuring adequate intensity to reach the algal growth, a high light consumption efficiency can be attained.

3.3. Potential of Algal strain for biofuel production

The previous analysis revealed that the strains Cv and NS are the most commonly utilized in wastewater treatment (WwT) processes. Given their widespread application and effectiveness, these strains have been selected to optimize their potential for biofuel production. This decision is based on their proven performance and potential adaptability to WwT and biofuel production processes, where their robust characteristics may enhance efficiency and yield. The next steps will involve detailed examination and optimization of biofuel production parameters specifically using these strains. It has been reported that NS strain tolerate moderate range of L_I ($50\text{--}420 \mu\text{E m}^{-2} \text{s}^{-1}$) and can stand a wide range of $C_{c,g}$, 0.03–80 % (Almomani et al., 2019b; Liu et al., 2013; Watanabe and Fujii, 2016). The oil content in this strain has reported to be between (40–31.3 %) with triacylglycerol (TAG) content up to 21 % (Stepan et al., 2016; Li et al., 2014; Takagi et al., 2000; Jazzar et al., 2015b; Wahidin et al., 2013; Teo et al., 2014). The lipid content of Cv has reported to be as high as 58 % (Kumar et al., 2010) and tolerant to $C_{c,g}$, as high as 15 % and temperatures of 30°C or more, and can grow under various cultivation modes. Fig. 3 presents the most important

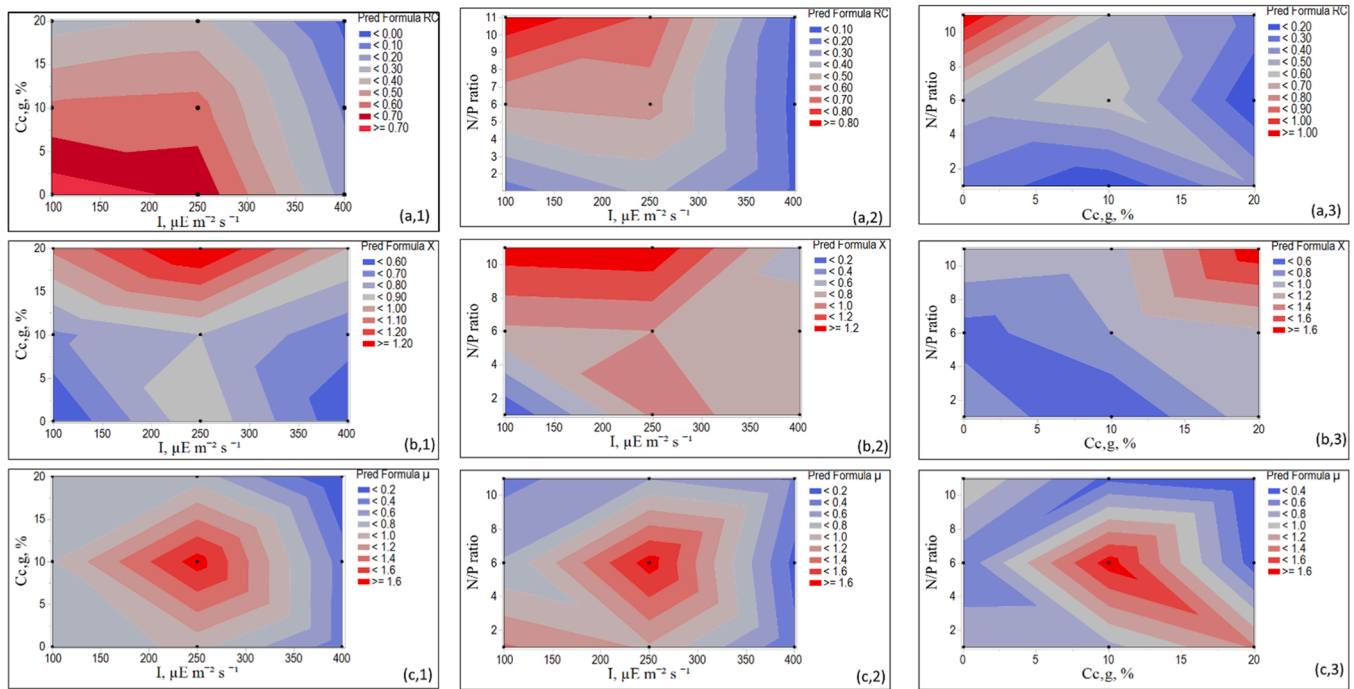


Fig. 4. Contour plots and prediction formula for R_C (a), X (b) and μ (c) as a function key input parameters of nitrogen to phosphorus ratio ($R_{N/P}$), CO_2 gas concentration ($C_{c,g}$) and light intensity (I_L).

growth parameters (μ , X_{max} , and HRT) for *C.v* and *Ns*. Results reveal that *Cv* has X and μ in the range 1.1 – $2.75 g L^{-1}$ and $0.13 d^{-1}$ – $0.66 d^{-1}$, with a maximum ($X_{max} \approx 2.75 g L^{-1}$ and $\mu_{max} \approx 0.66 d^{-1}$) achieved at HRT of 2 days. While the recorded data for *NS* were in the range 0.023 – $2.10 g L^{-1}$ and $0.13 d^{-1}$ – $0.66 d^{-1}$, with the maximum achieved at ($X_{max} \approx 2.10 g L^{-1}$ and $\mu_{max} \approx 2.7 d^{-1}$) and HRT of $2.7 d^{-1}$.

3.4. Optimum biofuel production (BY_p) synchronized WwT

The different input parameters and growth conditions of *Cv* and *NS* were used in the developed biokinetic. presents the contour plots for R_{CO2} , X , and μ as a function of these parameters. The obtained equations from this modelling process were shown in Table 3. The RMS, RMSE and $PV \leq 0.05$ were used a judgement factors for the significance of the current statistical analysis, whereas $PV > 0.05$ were considered insignificant and rejected. Table 3 shows the obtained formulas can successfully predict R_C , X , and μ based on the input values of $R_{N/P}$, $C_{c,g}$ and I_L with RMS value > 0.93 , maximum RSME ≤ 0.27 , and $PV \leq 0.05$. Positive and negative suggest the existence of optimum values under the studied conditions as presented in our previous work (Almomani, 2019, 2020). The two-dimensional (2D) response surface contours (Fig. 4a, b, and c) were created based on the computed quadratic equations. The actual experimental response data values appear on contour plots as discrete data points, added to clarify and compare the fit with generated model. The obtained results in Fig. 4a1, a2, a3 revealed that the R_{CO2} increased from 0.1 to $\geq 0.7 g L^{-1} d^{-1}$ when I_L and $C_{c,g}$ increased from 100 to $250 \mu E m^{-2} s^{-1}$ and 0.03 – 6% respectively. Higher I_L and $C_{c,g}$ ($300 \mu E m^{-2} s^{-1}$ and 11%) decreased the R_{CO2} slightly to $\leq 0.7 g L^{-1} d^{-1}$. The obtained R_{CO2} at $R_{N/P}$ from 10 to 11 was $\geq 0.8 g L^{-1} d^{-1}$, while for $R_{N/P}$ between 8.5 and 10 decreased R_{CO2} slightly to $\leq 0.8 g L^{-1} d^{-1}$. A maximum value of the $R_{CO2} \geq 1 g L^{-1} d^{-1}$ was reported for the operation conditions at $C_{c,g} 2\%$ and $R_{N/P}$ between 10 and 11 . It has been observed that $C_{c,g}$ of at least 1% needs to be provided to sustain the algal biomass. A maximum $X \geq 1.2 g L^{-1}$ was obtained at I_L and $C_{c,g}$ of 200 – $300 \mu E m^{-2} s^{-1}$ and 15 – 20% respectively. The optimum $R_{N/P}$ ratio that satisfied higher growth rate $\geq 1.2 g L^{-1}$ in the range 8 – 11 . A value of $1.6 g L^{-1}$ of X was achieved at $R_{N/P}$ and $C_{c,g}$ of 9 – 11 and 15 – 20% as shown in Fig.4b1,

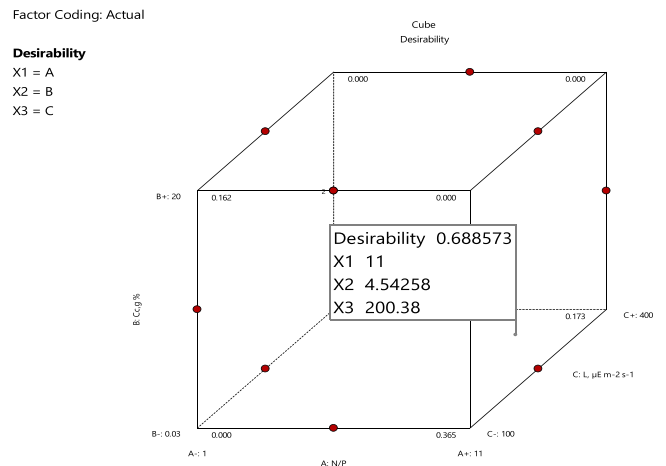


Fig. 5. Desirability cube for the selected responses.

b2, b3. A maximum $\mu \geq 1.6 d^{-1}$ at $C_{c,g}$ and I_L of 7.5 – 13.5% and 200 – $300 \mu E m^{-2} s^{-1}$, respectively, while optimum $R_{N/P}$ ratio between 4 and 7.5 increased $\mu \geq 1.6 g L^{-1}$. The combination influence of $R_{N/P}$ and $C_{c,g}$; 3 – 7.5 and 8 – 15% respectively increased $\mu \geq 1.6$ (Fig.4c1, c2,c3). The study noted that the initial $R_{N/P}$ ratio significantly affects both microalgae growth and the removal capacities of nitrogen (N) and phosphorus (P), with the optimal $R_{N/P}$ ratio identified as 10 (Alketife et al., 2017). High light irradiance can harm algal photosynthetic units, causing photo-inhibition (Alketife et al., 2017). As solar irradiance drops, photosynthesis-active radiation levels drop, reducing photosynthetic activity and CO_2 fixation rates (Alketife et al., 2017). R_{CO2} demonstrates an increase with rising $C_{c,g}$, ranging from 0.03% to 5% , and also exhibits an increase with I_L up to $100 \mu E m^{-2} s^{-1}$ (Al Ketife et al., 2017). The optimum conditions obtained in the current study are; $I_L = 100$ – $270 \mu E m^{-2} s^{-1}$, $C_{c,g} = 0.03$ – 15% , $R_{N/P} = 3.5$ – 11 corresponding to a maximum response values of $R_{CO2} = 1 g L^{-1} d^{-1}$, $X = 1.6 g L^{-1}$, and $\mu = 1.6 d^{-1}$.

Table 4Predicted formulas for R_{CO_2} , X , and μ .

Predicted formula	Equation	RMS	RMSE	PV-value
$R_{CO_2}, g L^{-1} d^{-1} =$	$-0.33 + 3.6 \times 10^{-3} C_{c,g} + 4.5 \times 10^{-3} L + 1.5 \times 10^{-1} N/P - 2.5 \times 10^{-4} C_{c,g}^2 - 1 \times 10^{-5} L^2 - 4.9 \times 10^{-4} N/P^2 + 5.3 \times 10^{-5} C_{c,g} L - 5 \times 10^{-3} C_{c,g} N/P - 2.4 \times 10^{-4} L N/P$	0.97	0.092	0.0022
$X, g L^{-1} =$	$-0.7 - 2.7 \times 10^{-2} C_{c,g} + 8.7 \times 10^{-3} L + 9.6 \times 10^{-2} N/P + 1.9 \times 10^{-3} C_{c,g}^2 - 1.1 \times 10^{-5} L^2 + 4.35 \times 10^{-3} N/P^2 - 2.5 \times 10^{-5} C_{c,g} L + 3 \times 10^{-3} C_{c,g} N/P - 5 \times 10^{-4} L N/P$	0.97	0.1	0.003
$\mu, d^{-1} =$	$-0.85 + 1.52 \times 10^{-1} C_{c,g} + 1.42 \times 10^{-2} L + 1.23 \times 10^{-1} N/P - 5.44 \times 10^{-3} C_{c,g}^2 - 3.47 \times 10^{-5} L^2 - 1.5 \times 10^{-2} N/P^2 - 4.9 \times 10^{-5} C_{c,g} L - 6.6 \times 10^{-3} C_{c,g} N/P + 3.3 \times 10^{-4} L N/P$	0.93	0.27	0.02

3.5. Disability function for the selected reponses

The desirability functions of the input parameters ($R_{N/P}$, $C_{c,g}$ and L_I) were used to identify desirable input under optimum conditions. Fig. 5 shows the obtained desirability is 0.68, with optimum values of R_{C_i} , X achieved at $R_{N/P}$, $C_{c,g}$, and L_I values of 11, 4.54 % and $200 \mu E m^{-2} s^{-1}$, respectively.

3.6. Maximizing the biofuel production and nutrients removal

Algal hydrothermal liquefaction (HTL) was selected for downstream

process of microalgae cultivation technology (MCT) and biofuel yield (B_Y , %) (Al Ketife et al., 2020; Ross et al., 2010; Toor et al., 2011). The HLT process input parameters for both algal strains (Cv and NS) were; 200 bar, $340^\circ C$, and resident time of 3–10 mint, algae concentration of 20 %, 14 % lipids content for Cv and 35 % for NS (See Table 4). The $\%B_Y$ obtained out of HTL process was 40 % and 49 % for Cv and NS respectively. The maximum energy content obtained for the biofuel was $24 MJ kg^{-1}$ and $39.1 MJ kg^{-1}$ for Cv and NS respectively. The B_{Yp} was correlated to algae input growth parameters ($C_{c,g}$, L_I , and $R_{N/P}$) as shown in Fig. 6. Under hydrodynamic retention time (HRT) and the initial biomass concentration of 2 d and 10 % respectively, the $B_{Yp \max}$ of $1.47 g L^{-1} d^{-1}$ was achieved optimum MCT conditions of $C_{c,g}$ (4–16 %); L_I (200 – $250 \mu E m^{-2} s^{-1}$), and N/P (8–11). It is worth to mention that the reported B_Y is a function of algal bio-composition, and therefore the B_{Yp} values were $0.58 g L^{-1} d^{-1}$ for Cv and $0.72 g L^{-1} d^{-1}$ for NS respectively. The results obtained in this study are similar to those documented in the literature (Tables 1–3).

3.7. Economic study

The economy of the FES used in optimization of biofuel production relies on the cost of ($C_{c,g}$ %, Algal stain (AS), Energy (E), and nutrients (Nu) (see Fig. 7). Table 5 summarizes the results of the economic study. The analysis includes the cost for CO_2 gas fixation which was determined to be \$0.026 for Ns and \$0.023 for Cv, the cost of cultivating microalgae which was determined to be \$0.54/kg, the cost of the HTL of \$0.016/kg, and the output costs are \$0.46 for Ns and \$0.23 for Cv, the cost of required nutrients is \$0.026/kg, and the output costs for nutrients uptake are \$0.113 for Ns and \$0.0683 for Cv. The obtained results revealed that NS cost out (\$0.49) is higher than Cv (0.25) by 1.94 folds complying

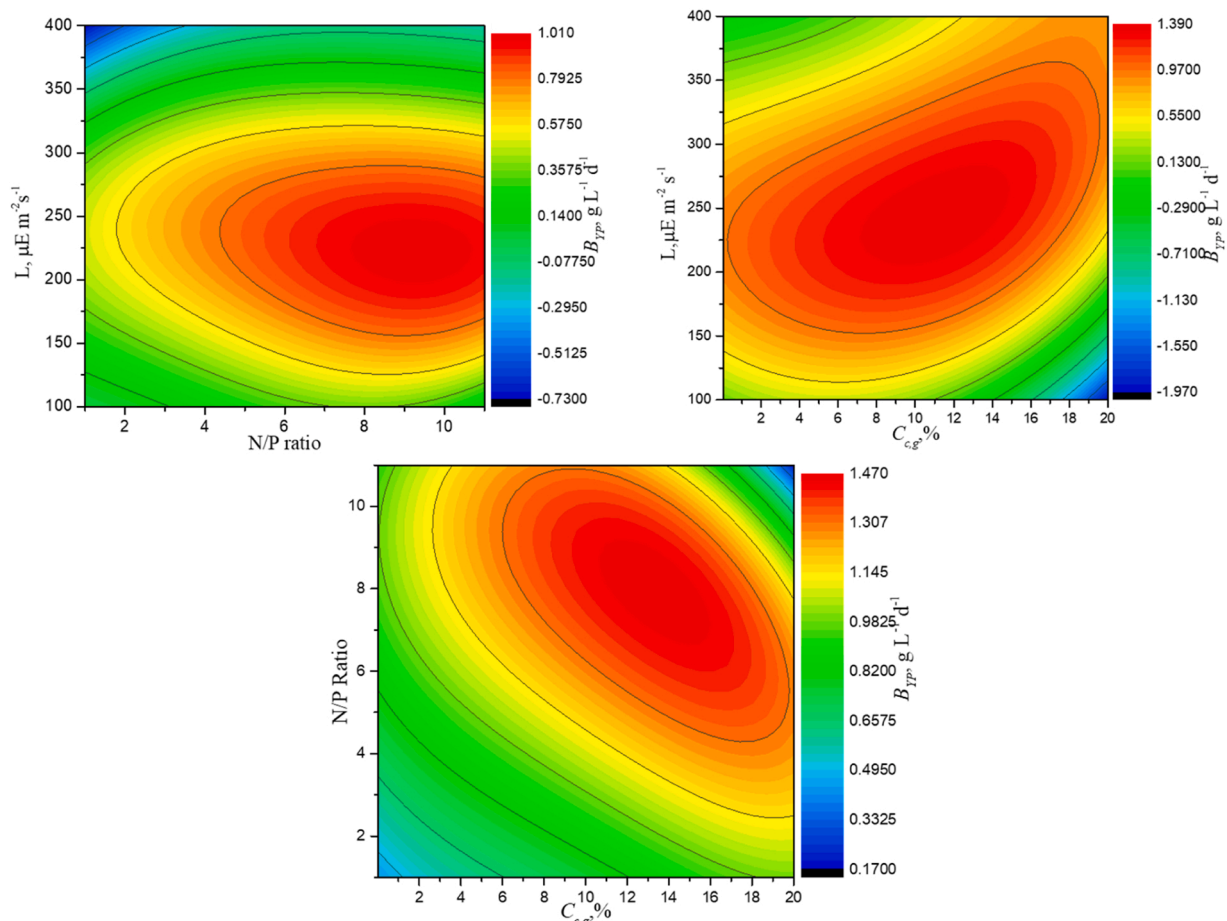


Fig. 6. Counter profile for Biofuel production (B_{Yp} , $g L^{-1} d^{-1}$) with respect to $C_{c,g}$ %, L_I and $R_{N/P}$.

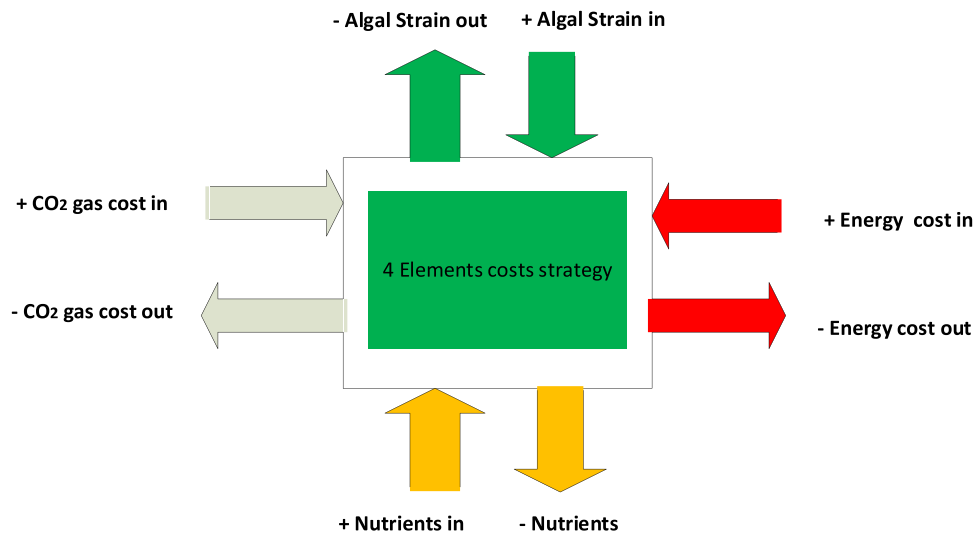


Fig. 7. Cost balanced of the four elements strategy (FES).

Table 5
Biocrude yield and HTL operation conditions for Cv and NS (Lammers et al., 2017).

Parameters	<i>Chlorella vulgaris</i> (Cv)	<i>Nannochloris</i> (NS)
Pressure (bar)	200	140–200
Temperature (C)	340	350
Algal concentration (wt%)	20 %	20 %
Resident Time (min)	3	1–10
Lipids content (wt%)	14 %	35 %
Biofuel Yield (B_Y , %)	40 %	49 %
$BY_{p,Bio}$ –oil HHV	24 MJ/Kg	39.1 MJ/kg

with the recorded figures for nutrients uptake rate and biofuel production for NS. The total cost required to produce 1 L of biofuel is 0.629 \$, and the required cost to cover this cost is 0.138 \$ and 0.375 \$ for NS and Cv respectively (Table 5).

4. Conclusion

Despite favourable potential of microalgae for biofuel and value-added products, challenges related to cost-effectiveness, scalability, and technological advancements in processing methods must be addressed. This study introduces a comprehensive methodology to optimize Microalgae-Based Biofuel Production, employing a four-element strategy that emphasizes crucial growth parameters such as algal strain, nutrient concentrations, energy inputs, and CO₂ concentrations. Optimizing these factors and assessing desirability, the study aims to determine optimal conditions for maximizing efficiency in the algal cultivation process. The study also involves a thorough investigation into the mechanism of nitrogen and phosphorus removal by various algal strains. Nitrogen to phosphorus ratio ($R_{N/P}$) emerges as a critical factor influencing efficient algal growth, emphasizing the importance of maintaining a proper balance. Saline strains exhibit higher CO₂ bio-fixation compared to freshwater strains. The study highlights the importance of operational factors such as temperature, CO₂ concentration, algal strain, and nutrients in optimizing algae growth. A correlation between CO₂ fixation and light intensity was established confirming the

Table 6
FES economic calculations, \$ Kg^{-a}.

Element	Cv	Ns	Notes	References
+ CO ₂ gas cost in	0.02		Cost of CO ₂ gas delivered, calculated from current study based on 52 % carbon content in dry biomass.	(Al Ketife et al., 2020, 2019)
- CO ₂ gas cost out	0.023	0.026	Cost of CO ₂ gas fixation, estimated current study.	
+ Algal strain in	0.54		Cost of microalgae cultivation, calculated from current study including: APR ^a mixing costs; cost of inoculum based on GWPsR ² reactors; staffing levels and local labour costs based Gulf Region pricing; and cost of full disinfections of required power of Ww of 35 Wh.m ⁻²	(Al Ketife et al., 2020, 2019; Babar, 2020; Hoffman et al., 2017; Guccione et al., 2014; Rodolfi et al., 2009)
- Algal strain out	0		non	
+ Energy cost in	0.016		Calculated from current study including: Cost of HTL process, assume 50 % conversion factor; and energy required for cells harvesting.	(Al Ketife et al., 2020, 2019; Malkin and Fork, 1996; Pilon et al., 2011)
- Energy cost out	0.23	0.46	Cost of Biofuel yield, estimated current study.	
+ Nutrients in	0.026		Cost of nutrients required, calculated from current study encompassing: Ww pumping cost; and cost of abiotic including filtration.	(Al Ketife et al., 2020, 2019; Salih, 2011; Sayre, 2010)
- Nutrients out	0.0683	0.113	Cost of Nutrients uptake, estimated current study.	
Total cost in	0.62951			
Total cost out, NS	0.25	0.49		
Total cost out NS/ Cv	1.94			

^a APR: algal pond reactor;²GWPsR: green wall panels reactors.

effect of light on the CO₂ fixation, nutrient removal, and lipid accumulation. The integration of the four-element strategy, coupled with advanced statistical and biokinetic approaches, demonstrates promising results in achieving efficient wastewater treatment and high biofuel productivity. The results indicate that the saline strain (Ns) demonstrates superior nutrient removal capacities and biomass production compared to the freshwater strain (Cv). For the studied strains (*Nannochloropsis* sp. (Ns) and *Chlorella vulgaris* (Cv)), results showed that optimal conditions for maximizing CO₂ bio-fixation rate (1 g L⁻¹ d⁻¹), biomass accumulation (6 g L⁻¹), and algal growth rate (1.6 d⁻¹) were L_I in the range 100–270 μE m⁻² s⁻¹, inlet C_{cg} in the range 0.03–15 % and R_{N/P} ranging from 3.5 to 11. Further optimization using the desirability function improved these metrics to 1.2 g L⁻¹ d⁻¹, 1.8 g L⁻¹, and 1.7 d⁻¹, respectively at an R_{N/P} = 11, C_{cg} = 4.5 %, and L_I = 200 μE m⁻² s⁻¹. The economic feasibility analysis reveals higher costs associated with the saline strain (Ns), attributed to its elevated nutrient uptake rate and biofuel production. Ns demonstrated superior nutrient uptake rates and biofuel yield production (BY_P) compared to *Chlorella vulgaris* (Cv), with a significantly lower production cost of \$0.629/ L.

CRedit authorship contribution statement

Fars Almomani: Writing – original draft, Supervision, Investigation, Conceptualization. **Ahmed M.D. Al Ketife:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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